

Mark Scheme for 2011 Prelim PU2 H1 Biology Paper 1

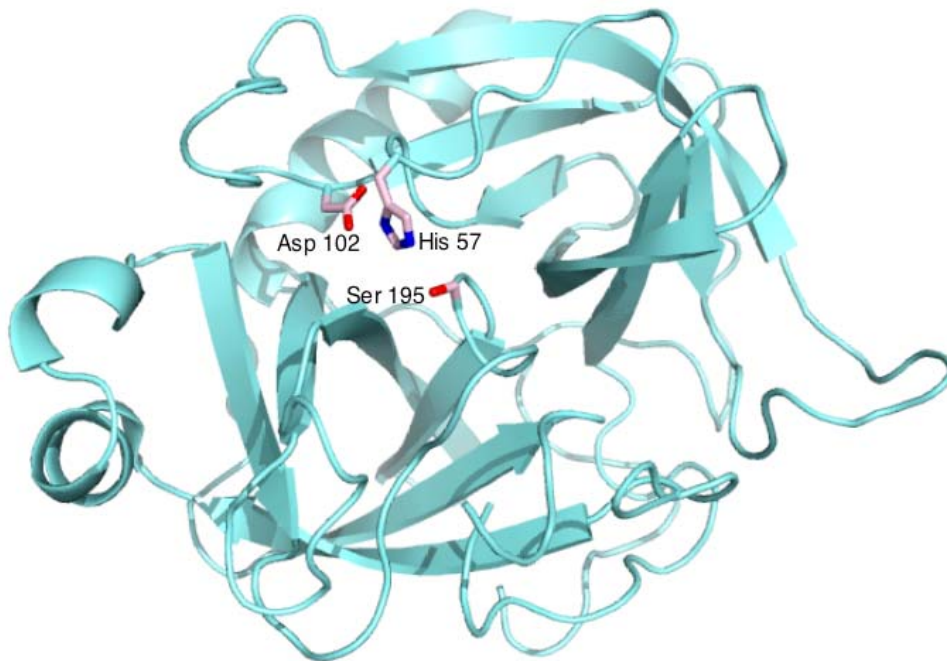
1	2	3	4	5	6	7	8	9	10
A	B	A	D	B	B	A	A	D	C
11	12	13	14	15	16	17	18	19	20
B	C	C	C	C	B	C	A	D	B
21	22	23	24	25	26	27	28	29	30
A	B	B	D	C	D	C	A	B	D

Mark Scheme for 2011 Prelim PU2 H1 Biology Paper 2**Section A**

Answer **all** questions in this section.

1. Chymotrypsin catalyses the hydrolysis of peptide bonds of proteins. It is made up of 241 amino acids. **Fig 1.1** shows the arrangement of three amino acids being in different positions in the amino acid sequence. The active site of chymotrypsin is located in a slight depression on one side of the molecule. Three amino acids form the active site. Even though these three amino acids are some distance apart on the polypeptide chain but close together in the active site.

1	Histidine 57	Aspartic acid 102	Serine 195	241

Fig 1.1**Fig 1.2**

- (a) Describe how amino acid residues at different positions in the protein may be brought together in the active site when the enzyme is synthesised.

- brought close together in the active site by the folding of polypeptide chain into a globular shape;
- amino acid sequence/primary structure determines the different chemical nature of R groups eg. ionic / polar / non-polar / acidic / basic(at least 2 examples);
- positions maintained by hydrophobic interactions, hydrogen bonds, ionic bonds and disulphide bonds(at least 2 examples);

.....

.....[2]

Fig. 1.3 shows the effects of increasing substrate concentration on the rate of a chymotrypsin-catalysed reaction at a temperature of 35°C and a constant pH.

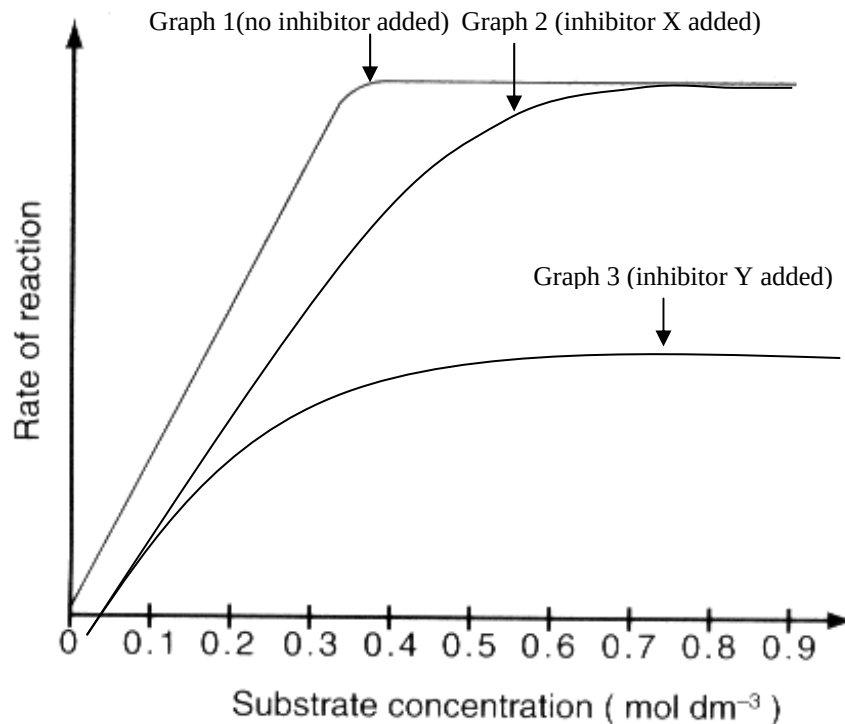


Fig 1.3

- (b) With reference to graph 1 in Fig 1.3, describe and explain the effect of substrate concentration on the rate of enzyme-catalysed reaction at high substrate concentration.

- At high substrate concentration further increase in substrate conc., no further increase in rate of reaction, ref. figures 0.37 to 0.9 mol dm^{-3} / beyond 0.37 mol dm^{-3} , all available active sites are occupied/bound by substrates / enzyme/trypsin (becoming)
- saturated with substrate;
- substrate is no longer limiting / enzyme concentration is limiting;
- ; for 1 mark and max 2 marks

.....[2]

- (c) Two different types of inhibitors, X and Y, were added and their results were shown in graph 2 and 3 respectively Describe two differences between inhibitor X and Y

- X is structurally similar to the substrate + Y is not structurally similar to the substrate;
- X binds reversibly to the active site of the enzyme + Y binds to a region other than the active site of the enzyme i.e. allosteric site;
- X competes with substrates for active site / do not render the enzymes as non-functional and can be re-used + Y cause a conformational change of the active site after binding to the allosteric site / renders the enzyme as non-functional;
- For X, inhibition can be overcome high substrate concentration + for Y, the rate of reaction does not reach the same V_{max} as that catalyzed by uninhibited enzyme at high substrate concentration;
- ; for 1 mark and max 2 marks

....[2]

Table 1.4 shows some mRNA codons and the corresponding amino acids for which they code. **Fig. 1.5** shows

- the sequence of 4 amino acids found in the middle section of the human chymotrypsin polypeptide
- the corresponding nucleotide bases in the DNA.

amino acid	abbreviation	mRNA codons					
glutamic acid	glu	GAA	GAG	–	–	–	–
phenylalanine	phe	UUU	UUC	–	–	–	–
lysine	lys	AAA	AAG	–	–	–	–
proline	pro	CCA	CCC	CCG	CCU	–	–
threonine	thr	ACA	ACC	ACG	ACU	–	–
valine	val	GUA	GUC	GUG	GUU	–	–
cysteine	cys	UGC	UGU	–	–	–	–
arginine	arg	CGC	CGA	CGU	CGG	AGA	AGG

Table 1.4

Amino acid Sequence	Arg	Cys	Lys	Glu
mRNA sequence				
Coding DNA sequence	GCA	ACA	TTT	CTT

Fig. 1.5

(d)

(i) Use the information in **Table 1.4** to complete the nucleotide sequences for the mRNA shown in **Fig. 1.5**. Write your answer on **Fig. 1.5**. [1]

(ii) A mutation arises and results in a change in the nucleotide sequence shown in **Fig. 1.5** to **GCAACAATTCTT**. Identify the type of mutation that has occurred.

Nonsense mutation ; for 1 mark and max 1 mark	[1]
--	----------

(iii) Describe the effect of the mutation.

Change the mRNA codon from AAA which codes for Isoleucine to stop codon UAA;
 Lead to early/premature termination of translation/polypeptide chain produced is shorter than normal;
 Produced a truncated protein, which cannot fold into its functional/proper conformation/shape;

; for 1 mark and max 2 marks

.....

.....

.....

.....[2]

[Total:10]

2. Fig.2.1 shows two organelles that are found in a liver cell.

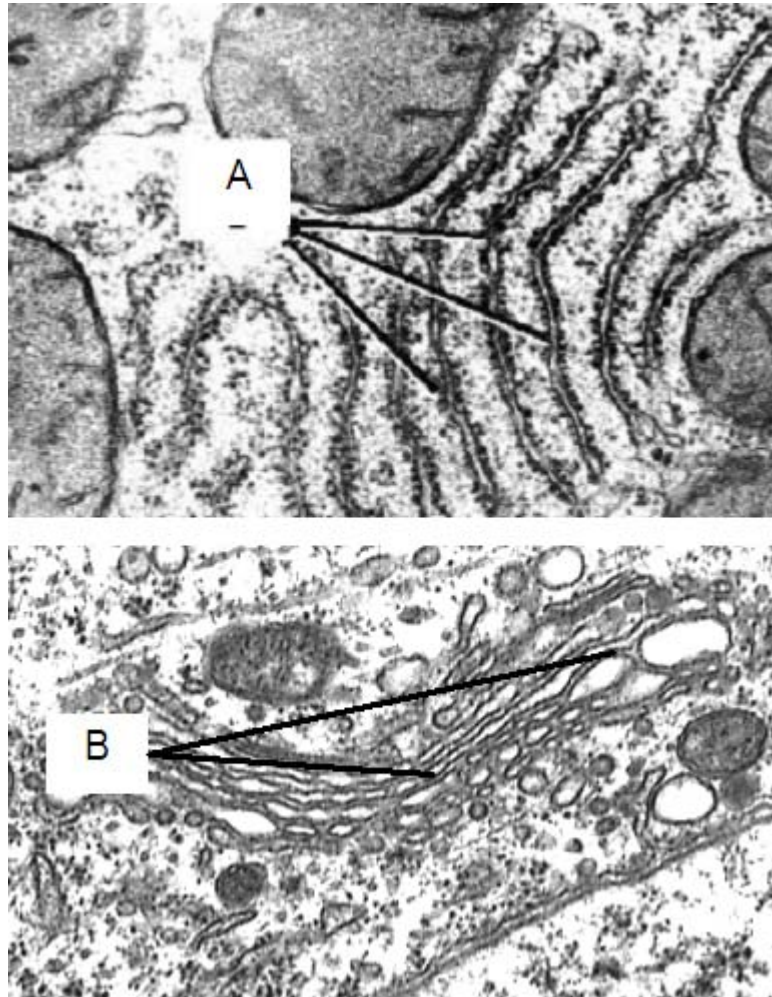


Fig. 2.1

- (a) Identify A and B in Fig 2.1.

A: A: rough endoplasmic reticulum;
B: Golgi apparatus;
B: ; for 1 mark and max 1 mark[1]

- (b) Describe the function of A and B with relation to the synthesis of a protein found outside the cell.

- Proteins produced by ribosomes on the RER;
Modification of protein in RER + vesicles containing protein bud off from the RER as transport vesicles;
- Transport vesicles travel to the Golgi apparatus + Golgi apparatus sort, package and further modification of proteins take place e.g adding carbohydrate chain;
- Modified proteins travelled through cisternae of Golgi apparatus + secretory vesicle pinches off (at the *trans* face);
- Secretory vesicles fuse with cell surface membrane and protein is released outside the cell by exocytosis
- ; for 1 mark and max 3 marks

.....
.....[3]

Fig 2.2 shows a diagram of a ribosome carrying out protein translation.

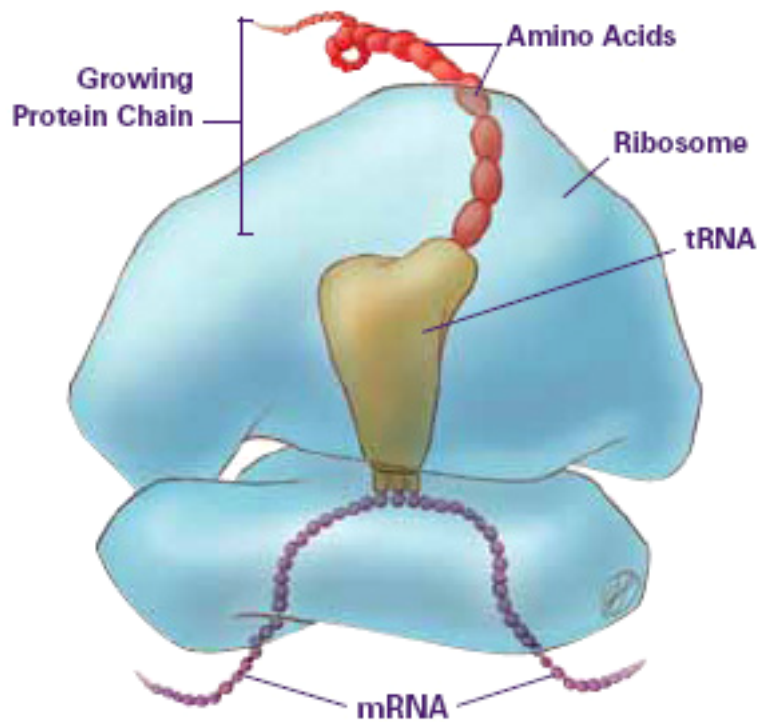


Fig. 2.2

(c)

(i) Explain how tRNA binds to mRNA

One terminal (base sequence of the tRNA)/anti-codon loop bears the anti-codon that is complementary to the codon found on mRNA;
The anticodon and the codon on the mRNA bind by complementary base pairing: A-U, C-G;
; for 1 mark and max 2 marks

.....

.....

.....

.....

.....[2]

(ii) Describe how the structure of the ribosome is adapted for translation.

Large subunit contains peptidyl transferase activity for formation of peptide bonds in between amino acids;
Small subunit for attachment of mRNA;
Large and small ribosomal subunits assemble to form Peptidyl, aminoacyl and exit sites;
Peptidyl site contains the tRNA with the polypeptide chain attached + Aminoacyl site is for binding of the new aminoacyl tRNA complex;
Exit site for tRNA to be released from ribosome;

; for 1 mark and max 2 marks

.....[2]

Fig. 2.3 shows two diagrammatically chromosome from two cells from the cell organism undergoing different types of nuclear division.

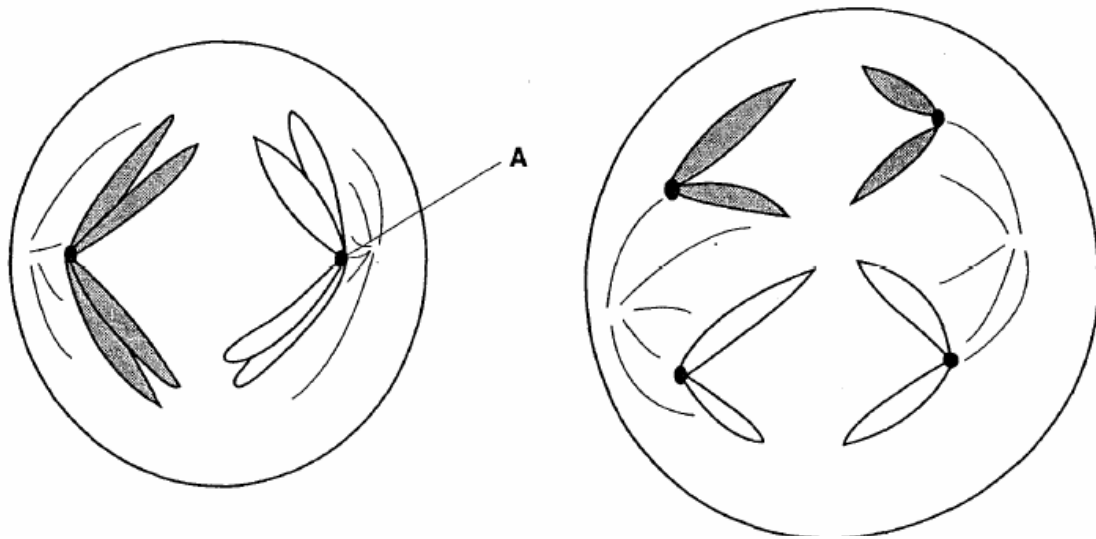


Fig. 2.3

(d) With reference to Fig. 2.3, state two functions of structure A

- attachment of spindle fibres on kinetochore found on centromere during metaphase;
- allows for equal separation of genetic material to daughter cells during anaphase;
- ; for 1 mark and max 3 marks

.....

.....

.....[2]

[Total: 10]

3. Two heterozygous pea plants were grown from round, yellow seeds. These plants were crossed and the seeds produced were recorded.

325 round, yellow
 100 wrinkled, yellow
 90 round, green
 30 wrinkled, green

The expected phenotypic ratio for this cross is 9:3:3:1

- (a) Provide reasons to explain why the observed phenotypic ratio is not 9:3:3:1

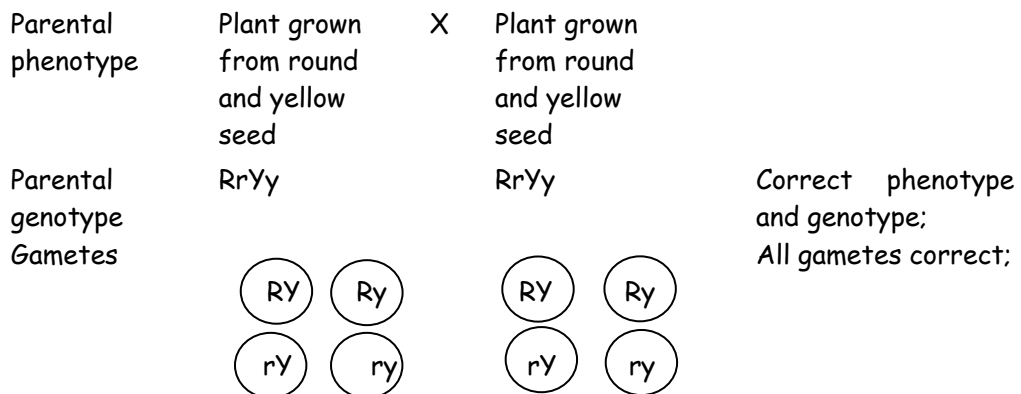
Small sample size or sampling error;

.....[1]

- (b) Using suitable symbols, construct a genetic diagram to explain the results

Symbols:

Let R represents the dominant allele for round seed
 r represents the recessive allele for wrinkled seed
 Y represents the dominant allele for yellow seed
 y represent the recessive allele for green seed



Gametes	RY	Ry	rY	ry
RY	RRYY round, yellow	RRYy round, yellow	RrYy round, yellow	RrYy round, yellow

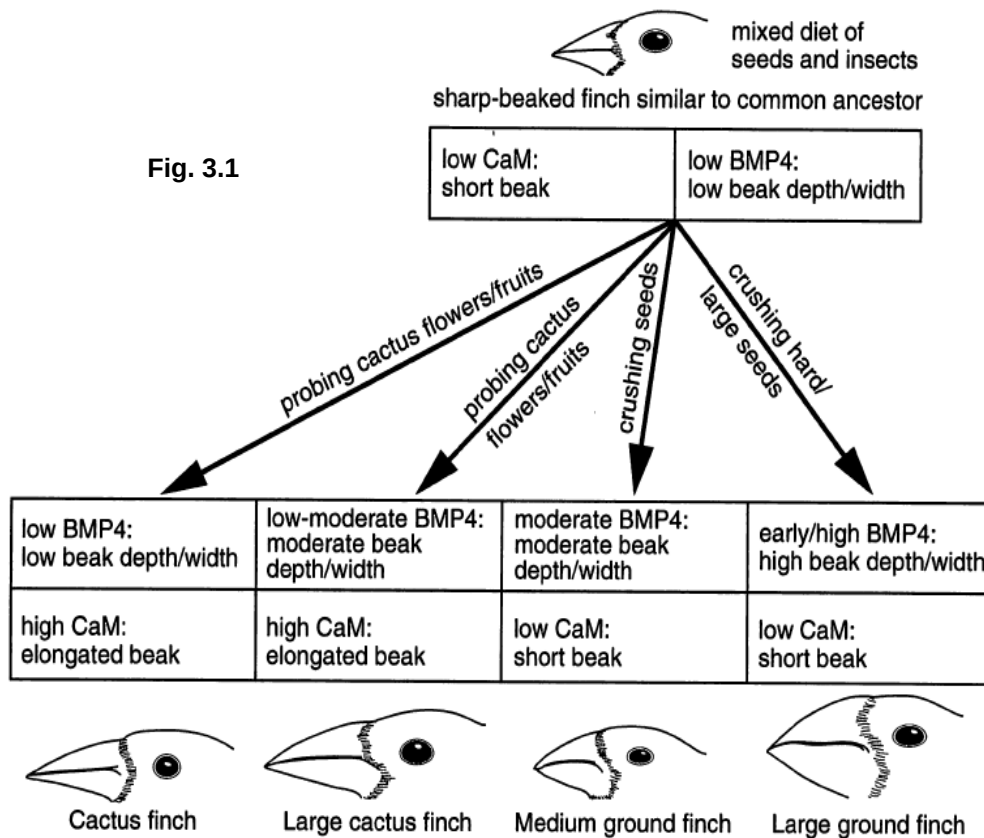
Ry	RRYy round, yellow	RRyy round, green	RrYy round, yellow	Rryy round, green
rY	RrYY round, yellow	RrYy round, yellow	rrYY wrinkled, yellow	rrYy wrinkled, yellow
ry	RrYy round, yellow	Rryy round, green	rrYy wrinkled, yellow	rryy wrinkled, green
Punette square to show correct fusion of gametes;				
Phenotype	Round, yellow	Round, green	Wrinkled, yellow	Wrinkled, green
Offspring phenotypic ratio	9 : 3	3 : 1	3 : 1	1
Correct phenotypic ratio;				

; for 1 mark and max 3 marks

[3]

Evolutionary changes in the beaks of Darwin's finches have been instrumental in the adaptive radiation of these birds in the Galapagos Islands. There are a dozen species that are closely related to one another showing remarkable differences in beak shape and size, and in eating habits. They have all arisen from a common ancestor, but have evolved to exploit different food sources. The precise dimensions (length depth and width) of each species' beak are crucial to their survival, and fluctuations in the environment lead to selection that changes the relative success of birds with various beak shapes. The molecular basis for variation in beak size and shape appears to be due to two genes: the gene for **bone morphogenetic protein 4** (*BMP4*) that encodes a cell-cell signaling protein that is expressed more during the embryonic development of the deep and wide beaks of ground finches than during the development of narrower beaks. the gene for **calmodulin** (*CaM*) that encodes a calcium signaling protein that is expressed at much higher levels in the long and pointed beaks of cactus finch embryos.

Fig. 3.1 shows how these two genes interact to produce variation in beak size and shape.



- (c) Describe two environmental factors that drives the natural selection of finches in the Galapagos island

• **Type** of food, any two from cactus flowers, fruits, seeds and hard and large seeds;
• **Availability** of food, **fluctuations in environment** leading to variation in supply of food during different seasons;
• **Competition** for food, **cactus finch competing with large cactus finch** or **medium ground finch competing with large ground finch**;
• ; for 1 mark, max is 2 marks

.....[2]

- (d) Explain how the molecular evidence described supports Darwin's theory of natural selection

Most offspring **die before they reach reproductive age** as a result of **selection pressure** exerted by the environment like type, availability and competition for food;
Individuals in a population are **genetically different** from one another, hence there is **variation within a species**;
BMP4 and CaM being present in the ancestor and that there are different levels of gene expression which in turn leads to **variation**;
Finches that can feed with the features/characteristics **best suited for the environment** (or genotypes that express phenotypes) are **more likely to survive to reproductive age**;
Successful finches **produce offspring** with characteristics or features similar to themselves as the offspring **inherit the favourable genes** from the parents;
Hence the number of individuals with favourable genes will **predominate in a population** / the **frequency of certain alleles / genotypes will increase**;

: for 1 mark. max is 3 marks

.....[3]

[Total: 9]

4. The polymerase chain reaction (PCR) is a method of obtaining large amounts of DNA from a small initial sample. **Fig. 4.1** shows the main stages in the polymerase chain reaction.

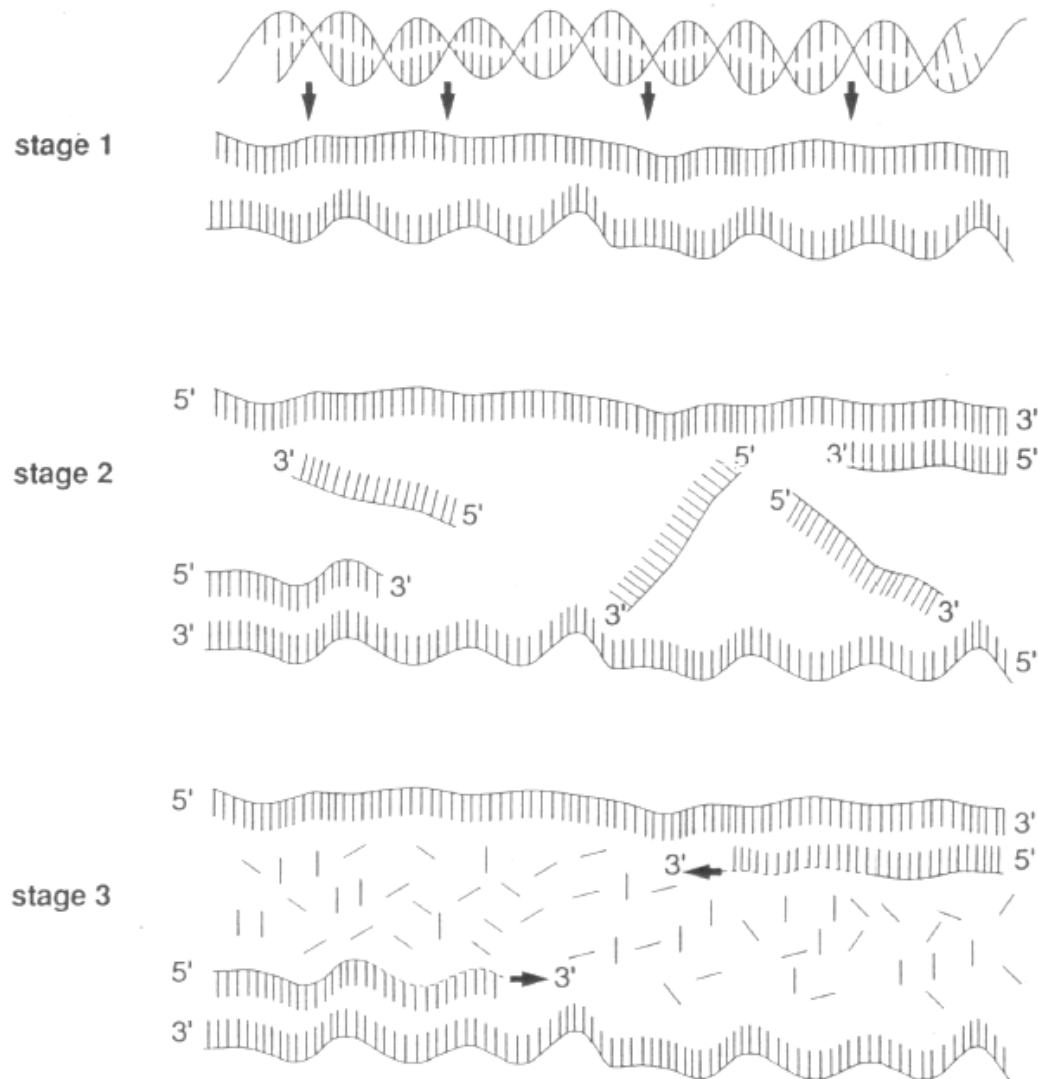


Fig. 4.1

(a)

- (i) Explain why DNA synthesis takes place in opposite directions on the two separated strands of the double helix.

Template DNA strands are antiparallel;
 Taq/DNA polymerase only adds nucleotides to the free 3' end of a growing DNA strand /
 synthesizes new strands in the 5' – 3' direction;
 ; for 1 mark and max 2 marks

.....

.....

.....[2]

- (ii) With reference to **Fig. 4.1**, briefly describe what is happening at each of the three stages

Stage 1 is denaturation, ds DNA is heated to 94-96 degree Celsius;
ds DNA separate into 2 strands;
; for 1 mark and max 1 mark

Stage 2 is annealing, temperature lower to 45-50 degree Celsius;
To allow primers to anneal
; for 1 mark and max 1 mark

Stage 3 is extension, DNA polymerase extend the primer by adding nucleotides
based on complementary base pairing with template strand;
for 1 mark and max 1 mark

.....[1]

The cloned DNA and plasmids were first cut by the same restriction enzyme and then mixed together with DNA ligase. Several types of plasmids were formed. Some contained the human DNA in the centre of the gene coding for resistance to tetracycline. The different types of plasmids are shown in Fig. 4.2.

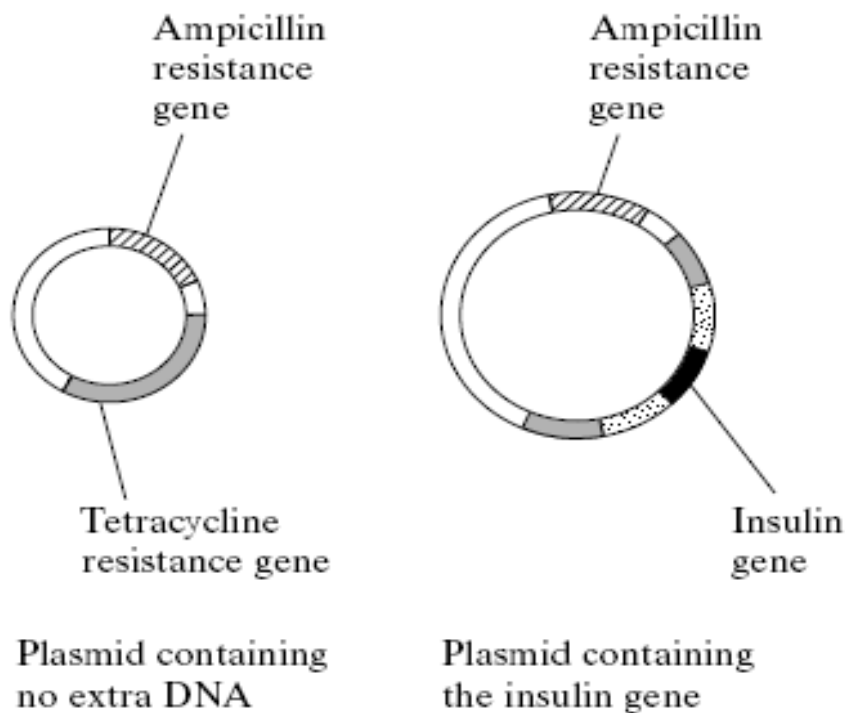


Figure 4.2

- (b) The plasmids are mixed with bacteria. Some bacteria took up the plasmids. Explain how it is possible to distinguish between bacteria which have taken up a plasmid with the insulin gene and those which have taken up a plasmid without any insulin gene

Ref to using antibiotic resistance selection as there are more than 1 antibiotic resistance gene in the plasmid;
Ref to using nutrient agar plates with ampicillin and tetracycline;
Ref to using bacteria that are transformed can grow on ampicillin medium as the plasmid confers the ability to breakdown ampicillin / resistance to ampicillin;
Ref to bacteria with the recombinant plasmid / has the gene insert cannot survive on tetracycline media as the tetracycline resistant gene is disrupted;

; for 1 mark and max 3 marks

.....[3]

Blood samples from 3 male adults, 1 female adult, and the baby were also taken and DNA extracted is amplified using PCR. Following PCR, the amplified DNA is cut into fragments with restriction enzymes. One of the loci on chromosome 7, **D7S820**, has many alleles, which differ from each other by the number of restriction sites present.

The amplified samples were loaded onto a gel and electrophoresis was performed to separate the fragments of DNA. They are separated by gel electrophoresis. The DNA is then treated with alkali and transferred onto a nitrocellulose membrane before radioactively labeled probes are added. The membrane is then washed and placed below a photographic film. The genetic fingerprint can be seen when the film is developed. A diagram of it is shown in **Fig. 4.3**

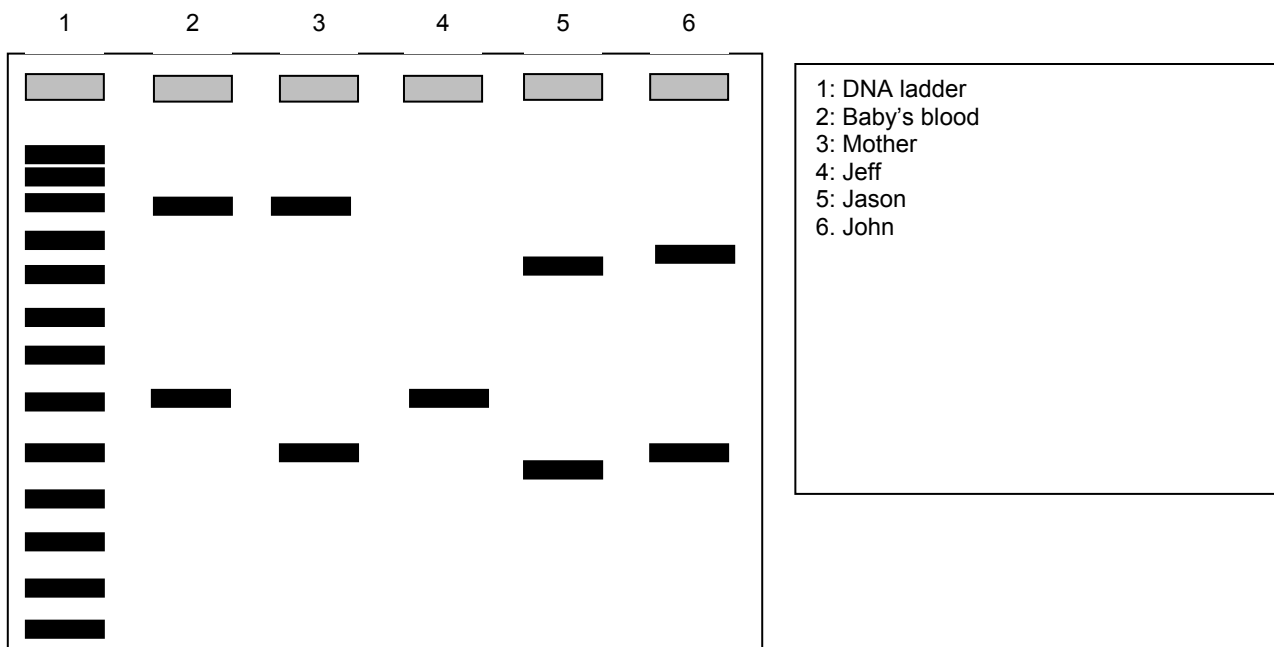


Fig. 4.3

(c)

(i) Account for the different position of the bands for each individual

Allele from each individual have different number of restriction sites, when cut by same restriction enzymes different number and size fragments are formed;
Fragments that have a larger molecular mass will move through the porous lattice slower because of greater resistance
Hence these fragments will be positioned further away from the wells as compared to the smaller molecular weight fragment;

; for 1 mark and max 2 marks

.....

.....[2]

16

(iii) Who is most likely to have fathered the child? Explain your choice.

Jeff + Position of the bands in lane 4 correspond to one of the band in lane 2;
for 1 mark and max 1 mark

.....

.....[1]

[Total: 11]

Section B

Answer one question

Write your answer to the question on the separate answer paper provided.
 Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
 Your answer must be in continuous prose, where appropriate.
 Your answer must be set out in sections (a), (b) etc, as indicated in the question.

Either

5.

- (a) Describe the different types of stem cells and explain their unique features and normal functions in a living organism [10]

Totipotent cells are cells that retain the complete set of genetic information and have the potential to become any cell type in the adult body, any cell of the extraembryonic membranes;
 Example of such is the fertilised egg and the first 4 cells produced by its cleavage;
 Pluripotent stem cells are cells that retain the complete set of genetic information have the potential to become any cell type in the adult body but not those of the extraembryonic membranes;
 Examples of such are Embryonic Stem (ES) Cells. These are isolated from the inner cell mass of the blastocyst;
 Multipotent stem cells are cells that retain the complete set of genetic information have the potential to differentiate into a limited number of cell types, are usually used to replace dead or damaged cells;
 Examples of such are adult stem cell, which is an undifferentiated cell found among differentiated cells in a tissue or organ;
 can renew and differentiate to yield major specialised cell types of tissue or organ;
 They exist in a very small number in each tissue and remain non-dividing for many years until they are activated by disease or tissue injury;
 Then they enter normal differentiation pathways to form specialized cell types of the tissue in which they reside;
 Therefore, the primary role of adult stem cells in living organism is to maintain and repair the tissue in which they are found;

; for 1 mark and max 6 marks

Unspecialized. There is an absence of tissue-specific structures that allow it to perform specialized functions;
 Able to proliferate, capable of continually renewing and dividing (through cell division) for long periods;
 Stem cells are able to proliferate (replicate many times), unlike normal cells;
 Capable of differentiating into specialized cells types under appropriate conditions;
 Stem cells are able to undergo differentiation (give rise to specialized cells), when triggered by intracellular and extracellular signals;

; for 1 mark, max 4 marks

(b) Describe the structure and organisation of DNA in eukaryotes.

[6]

2 polymers of deoxyribonucleotides chains twisted into a double helix;
 held by H- bonds between complementary bases, A=T, C≡G;
 2 nm diameter, 0.34nm/10 bp per complete turn;
 Anti-parallel, distinct polarity 3' end with free hydroxyl group at 3C and 5' end with phosphate group at 5C;
 DNA molecule associated with 8 histone proteins/octamer, formed nucleosome core + histones positively charged, formed ionic bonds with negatively charged DNA;
 nucleosome, linked by spacer/linker DNA forming 10nm 'beads-on-string' structure;
 further coiled into a solenoid forming a 30nm fibre;
 further looping in presence of scaffold proteins to form chromosomes during mitosis ;
 ;1 mark and max 6 marks

(c) Discuss how the proteins present in the membrane structure enables the membrane to be selectively permeable.

Proteins are either intrinsic (integral) or extrinsic (peripheral) interspersed in lipid bilayer;
 Intrinsic proteins span the membrane width and can serve as protein channels or protein carrier for movement of lipid insoluble substances;
 Intrinsic proteins can act as ion channels, carrier proteins protein pumps
 Ion channels will facilitated movement of substances across the membrane by facilitated diffusion;
 Protein carrier will undergo conformational change upon binding to a ligand, thus allowing movement of substances across the membrane;
 While protein pumps, with the aid of ATP, transport ions/ substances across the membrane against concentration gradient;
 ; for 1 mark, max 4 marks

[Total: 20]

6.

(a) Explain how the structure of chloroplast is adapted to its role in carrying out photosynthesis efficiently.

[8]

double membrane / chloroplast envelope isolated the photosynthetic reactions from the cytosol for optimal conditions / greater efficiency;
 membrane compartmentalizes/divide/separate chloroplast into thylakoid spaces and stroma;
 membrane that forms the thylakoid / grana increases the surface area;
 for attachment photosystems + for attachment of many electron carriers / electron transport chain (ETC) and enzymes such as ATP synthase;
 electron carriers are arranged in progressively lower energy level in the thylakoid membrane;
 the ordered arrangement of pigments and enzymes in the photosystems also helps with the efficient capture of light energy;
 the thylakoid membrane is impermeable to H^+ ;
 and this helps to maintain electrochemical gradient in the thylakoid space;
 to form proton gradient force for chemiosmosis;
 ATP synthase embedded in the thylakoid membrane are used in the production of ATP;
 stroma of chloroplast contains enzymes and other substance required for Calvin cycle;

;1 mark and max 8 marks

(b) Describe the process of oxidative phosphorylation in respiration.

[8]

Occur during aerobic respiration in the presence of oxygen;
 Oxidative phosphorylation produces 90% of the ATP generated by respiration / 34 out of 38 ATPs;
 As electrons are passed down the Electron Transport Chain, energy released is used to create a proton concentration gradient, across the inner membrane of mitochondria, needed for ATP synthesis;
 This creates the proton motive force;
 Hydrogen ions are passed back into the mitochondrial matrix, down a concentration gradient, through ATP synthase to synthesize ATP;
 Each reduced NAD molecule potentially produced three molecules of ATP;
 Each reduced FAD molecule potentially produced two molecules of ATP;
 Oxidative phosphorylation also allows the oxidation of NADH and $FADH_2$ / to regenerate NAD^+ and FAD;
 Oxygen act as a final electron acceptor in the Electron Transport Chain and maintain the flow of electrons along the Electron Transport Chain;
 Metabolic water is also produced through reduction of O_2 / when the electron is finally transferred to oxygen;

: 1 mark, max 8 marks

(c) Distinguish between Krebs cycle and Calvin cycle

[4]

Krebs cycle in mitochondrial matrix + Calvin cycle in stroma of chloroplast;
 Krebs cycle is catabolic process where acetyl CoA produced from glucose is broken down to produce ATP and NADH and $FADH_2$ + Calvin cycle is an anabolic process where glucose is formed through a series of reactions using CO_2 , and NADPH and ATP produced in the light reactions;
 In kerb cycle, CO_2 is released by oxidative decarboxylation + in calvin cycle, CO_2 is combined with ribulose biphosphate, catalysed by Rubisco;
 Coenzyme required in Krebs cycle is NAD^+ and FAD while in Calvin cycle, $NADP^+$ is required;
 In krebs cycle, ATP is synthesised by substrate level phosphorylation + in calvin cycle, ATP is used to convert 3-phosphoglycerate to glyceraldehyde 3-phosphate, and the regeneration of RuBP from glyceraldehyde 3-phosphate;

;1 mark and max 4 marks

End of Paper